

PSET 4

$$1(b) \frac{\ln 2 - \ln(1)}{c} = \frac{1}{c}$$

$$c > \frac{1}{\ln 2}$$

$$2(b) \quad 1 + \frac{n}{2} > \sqrt{1+n}$$

$$\text{at } x=0$$

$$1 > \sqrt{1}$$

at $n=0$

$$g(n) = 1 + \frac{n}{2} - (1+n)^{1/2}$$

$$\frac{1}{2}$$

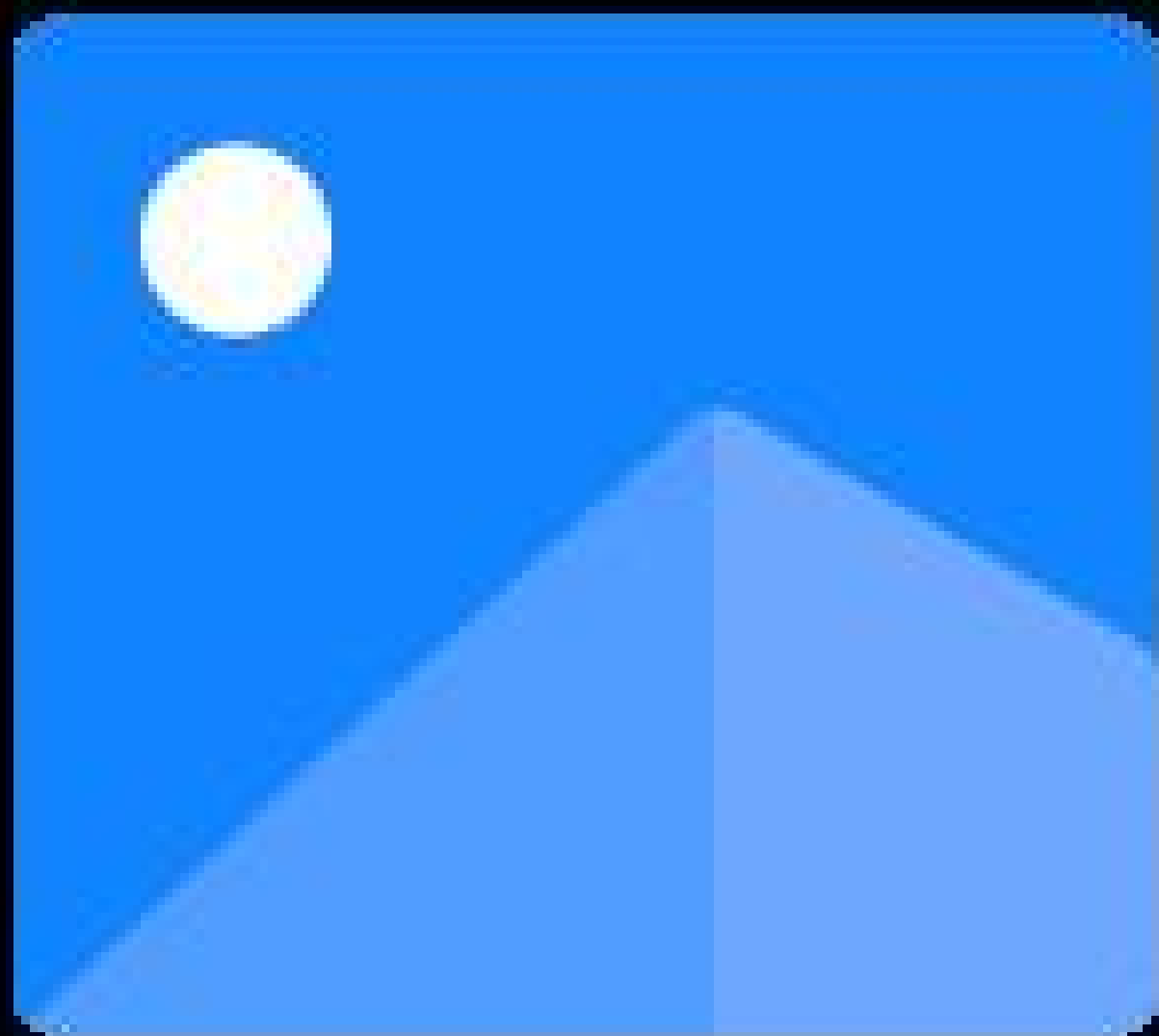
$$g'(n) = \frac{1}{2} - \frac{1}{2}(1+n)^{-1/2}$$

$$= \frac{1}{2} - \frac{1}{2\sqrt{1+n}}$$

$$\frac{1}{2\sqrt{1+n}} \text{ will be less than } \frac{1}{2}$$

for all values $n > 0$, so

$\frac{1}{2}$ - something smaller > 0



5. a) $f''(c)$ if p_1 is a point between $a < p_1 < b$
 where p_1 is the point where $f(b) = f(a)$, $f'(p_1) = 0$

if p_2 is a point between $b < p_2 < c$
 where p_2 is the point where $f(c) = f(b)$, $f'(p_2) = 0$

So, if we apply the theorem again there will be a point p between p_1 and p_2 , therefore between a and c , where $f''(p) = 0$

b) $f(x) = (x-a)(x-b)(x-c)$

We can do the same proof but a, b, c are the roots of the equation

6. a) $\frac{f(b) - f(a)}{b - a} = f'(c)$ if $b > a$

$f(b) - f(a) = f'(c) \cdot (b - a)$, $f'(c) > 0$, so +ve / +ve means +ve, therefore $f(b) - f(a) > 0$, therefore f is increasing

b) $f(b) - f(a) = f'(c) \cdot (b - a)$, if $f'(c) = 0$

$f(b) - f(a) = 0$ | Stays same i.e. $f(x)$ is a constant

$f(b) = f(a)$

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PART 11

$$1a) f(n) - f(0) = f'(c) \cdot (n - 0)$$

$$f(n) = f'(c)(n - 0), \text{ if } c > 0, \text{ then we know } f'(c) > 0, \therefore f(n) > 0$$

$$b) f(n) > n - \ln(1+n)$$

$$f(n) = n - \ln(1+n)$$

$$f(0) = 0$$

$$f'(n) = 1 - \frac{1}{1+n}, \text{ for value } n > 0, \text{ it will be}$$

something

be positive

$$c) \ln(1+n) > n - \frac{n^2}{2}$$

$$f(n) = \ln(1+n) - n + \frac{n^2}{2}$$

$$f(0) = 0 - 0 + 0 = 0$$

We know $n > 1/n$
 $0 < x$

$$\frac{1}{1+n} \geq (n+1)/n \quad (6)$$

$$f'(0) = \frac{1}{1+0} - 1 > 0$$

$$f''(n) = -(1+n)^{-2} + 1$$

$$f''(n) \geq 1 - \frac{1}{(1+n)^2}$$

$$= \frac{(1+n)^2 - 1}{(1+n)^2}$$

$$n \geq (n+1)/n \quad (9)$$

$$n \geq (n+1)/n$$

$$\geq 1 - \text{something smaller than } 1 \geq 0$$

$$f''(n) > 0, f'(n) > 0, f(n) > 0 \quad (n+1)/n \geq 1 + 1/n$$

$$\ln(1+n) \leq n - \frac{n^2}{2} + \frac{n^3}{3}$$

$$n - \frac{n^2}{2} + \frac{n^3}{3} \geq \ln(1+n)$$

$$f(n) = n - \frac{n^2}{2} + \frac{n^3}{3} - \ln(1+n)$$

$$f(0) = 0, f'(n) = 1 - n + n^2 - \frac{1}{1+n}$$

$$\frac{1}{1+n} = \frac{1}{1+n} + 1 - n + n^2$$

$$f''(n) = \left(\frac{1}{1+n} - 1 + n\right) + 2n$$

$$\ln(1+n) < n - \frac{n^2}{2} + \frac{n^3}{3} \quad \text{LO}$$

$$f'(0) = 0$$

$$f''(n) =$$

$$1 - n + (1+n)^{-2} + 2n$$

$$f''(0) = -2 + 2 = 0$$

$$f'''(n) = -(-2(1+n)^{-3} + 2)$$

$$= \frac{2}{(1+n)^3} - 2$$

2 will be greater than 2
 denominator

$\ln(1+x) \leq \frac{x^{\text{odd}}}{\text{odd}} - \dots + \dots$ $x \geq 0$
 $\ln(1+x) \geq \frac{x^{\text{even}}}{\text{even}} - \dots + \dots$ $-1 < x < 0$

e) $\ln(1+u) \leq u$ | From b) $\ln(1+t) \leq t$
 $\ln(1-v) \leq -v$ | $t = \frac{v}{1-v}$

$$\Rightarrow v \leq \ln\left(\frac{1}{1-v}\right) \text{ (using } \ln(1-x) \approx -x \text{)}$$

3A.1d) $d(e^{3u} \sin u) = (e^{3u} \cos u + \sin u e^{3u}) du$ $(u)'' = 2$
 $0 = (u)''$

1e) $\sqrt{n} + \sqrt{y} = 1$; $\sqrt{y} = 1 - \sqrt{n}$ $n \geq (n+1)/4$

$$\cos\left(1 + \frac{1}{2} \ln 2\right) + g^{\frac{1}{2}} = 1 \quad | \quad \text{(iii) at } \frac{e_2}{e} = \frac{\sin \alpha}{3} = 0$$

$$x + y = \frac{1}{2} x^{\frac{1}{2}} dx + \frac{1}{2} y^{\frac{1}{2}} dy = 0 \quad (-1) \text{ auf } -\frac{dx}{dy} = \frac{x^{-\frac{1}{2}}}{y^{-\frac{1}{2}}} = -\sqrt{\frac{y}{x}}$$

$$\frac{dy}{y} = \frac{-n^{-1/2} dn}{y^{-1/2}}$$

$$\phi(n+1) = \frac{1}{n+1} \left(n+1 - \frac{1}{n+1} \right) = -n^{-1/2} dn \cdot y^{1/2}$$

$$05 \quad = -n^{-1/2} dn \cdot (1 - \sqrt{n})$$

$$-n^{-1/2} + 1$$

$$\left(1 - \frac{1}{\sqrt{n}}\right) dn$$

$$2c) \int (2n^4 + 3n^3 + n + 8) dn$$

$$\frac{2n^5}{5} + \frac{3n^3}{3} + \frac{n^2}{2} + 8n + c$$

$$\frac{2}{5} n^5 + n^3 + \frac{1}{2} n^2 + 8n + c$$

$$c) \int \sqrt{8+9n} \, dn$$

$$\int (8+9n)^{1/2} \, dn$$

$$u = 8+9n \quad du = 9 \, dn$$

$$\int \frac{1}{9} u^{1/2} \, du$$

$$\int \frac{1}{9} u^{1/2} \, du$$

$$\frac{2}{9} u^{3/2} + c$$

$$\frac{2}{27} \sqrt{8+9n} + c$$

$$\frac{1}{6} u^{3/2} + c$$

$$\frac{2}{27} \sqrt{8+9n} + c$$

$$e) \int \frac{u}{\sqrt{8-2u^2}} du$$

$$v = 8-2u^2$$

$$dv = -4u du$$

$$\Rightarrow (8-2u^2)^{-1/2}$$

$$\frac{1}{2} (8-2u^2)^{-1/2} \cdot 4u^2 = \frac{1}{2} \int u^{1/2} dv$$

$$2u (8-2u^2)^{-1/2}$$

$$\frac{2u}{\sqrt{8-2u^2}}$$

$$\frac{1}{2} (8-2u^2)^{1/2} + C$$

$$\left(-\frac{1}{2} \right) \cdot \frac{1}{2} u^{1/2} + C$$

$$(8-2u^2)^{3/2}$$

$$3 \cdot \frac{1}{2} (8-2u^2)^{1/2} \cdot (-4u^2)$$

$$-6 (8-2u^2)$$

$$v = 8-2u^2$$

$$dv = -4u du$$

$$-\frac{1}{4} \int v^{-1/2} dv$$

$$-\frac{1}{4} \left(\frac{v^{1/2}}{1/2} \right) = -\frac{1}{2} v^{1/2} = -\frac{1}{2} (8-2u^2)^{1/2}$$

$$-\frac{1}{2}\sqrt{8-2n^2} + c$$

$$g) \int 7n^4 e^{n^5} dn$$

$$\frac{7}{5} e^{n^5} = e^{n^5} \cdot 5n^4$$

$$= 5n^4 e^{n^5} \cdot \frac{7}{5}$$

$$\frac{7}{5} e^{n^5} \cdot 5n^4$$

$$\frac{7}{5} e^{n^5} + c$$

$$i) \int \frac{dn}{3n+2} = \frac{1}{3} \ln(3n+2) + c$$

$$x \ln(3n+2) = \frac{1}{3n+2} \cdot 3 = \ln(n+5)$$

$$h) \int \frac{n}{n+5} dn = \ln(n+5) = \frac{1}{n+5} \cdot (n+5) = n$$

$$\int \frac{dv}{v} = \ln v$$

$$\ln n$$

$$k \int \frac{n}{n+5} dn$$

$$\frac{n+5-5}{n+5} = 1 - \frac{5}{n+5}$$

$$= n - 5 \ln(n+5) + C$$

$$3a) \int \sin(5n) dn$$

$$= -\cos 5n - \frac{1}{5} \cos(5n) + C$$

$$c) \int \cos^2 n \sin n dn ; u = \cos^2 n$$

$$= (1 - \sin^2 n) \sin n dn = -2 \cos n \sin n dn$$

$$\int (\sin n - \sin^3 n) dn = \left(-\cos n + \frac{1}{4} \cos^4 n \right) + C$$

$$= -\frac{1}{3} \cos^3 n + C = \left(-\frac{1}{3} \cos^3 n \right) + C$$

$$e) \int \sec^2(u/5) du$$

$$5 \tan^2(u/5) + C$$

$$g) \int \sec^9 u \tan u du$$

$$= \frac{1}{10} \sec^{10} u + C \quad \frac{1}{9} \sec^9 u + C$$

$$2b) \int \tan u \sec^2 u du = \frac{1}{2} \tan^2 u$$

$$y = \frac{1}{2} \tan^2 u$$

$$y' = 2 \cdot \frac{1}{2} \tan u \cdot \sec^2 u du$$

$$\tan u \sec^2 u du$$

$$f = \frac{1}{2} \sec^2 u$$

$$f' = \sec u \cdot \sec u \tan u du$$

$$= \sec^2 u \tan u du$$

equal, this is because when we do indefinite integrals there is a constant in the end and those constants will be different however those constants will evaluate with the functions will evaluate to the same family of functions

$$3F) 10) \frac{dy}{dx} = \frac{3}{\sqrt{y}}$$

$$dy = \frac{3}{\sqrt{y}} dx$$

$$dy \sqrt{y} = 3 dx$$

$$dy y^{1/2} = 3 dx$$

$$2/3 y^{3/2} = 3x + c$$

$$y^{3/2} = \frac{9x + c}{2}$$

$$y = \left(\frac{9x + c}{2} \right)^{2/3}$$

$$a = 3/2 c$$

$$d) \frac{dy}{dx} = ny^2$$

$$dy = ny^2 dx$$

$$\frac{dy}{y^2} = n dx$$

$$dy \cdot \frac{1}{y^2} = n dx$$

$$-1 y^{-1} = \frac{n^2}{2}$$

$$\frac{1}{y} = \frac{n^2}{2} + c$$

$$y^{-1} = -2 - c$$

$$y = \frac{-2}{n^2 + c}$$

$$2a) \frac{dy}{du} = 4ny$$

$$dy = 4ny \, du$$

$$\frac{dy}{y} = 4n \, du$$

$$\ln y = \frac{4n^2}{2} + c$$

$$y = e^{\frac{4n^2}{2} + c}$$

$$= e^{2n^2 + c}$$

$$= e^{2n^2} + 3 - e^2$$

$$= e^{2(3)^2} + 3 - e^2$$

$$= e^{12} + 3 - e^2$$

$$= e^{10} + 3$$

$$y(3) = e$$

$$y(1) = 3$$

$$3 = e^2 + c$$

$$3 - e^2 = c$$

$$\ln y = 2n^2 + c$$

$$\ln y(1) = 3$$

$$\ln(3) = 4c$$

$$c = \ln(3) - 2$$

$$\ln y = 2n^2 + (\ln 3 - 2)$$

$$\text{at } n=3$$

$$y = e^{18 + \ln 3 - 2} = 3e^{16}$$

$$A = C \cdot 0$$

$$ab = pb \cdot 0$$

$$A = C \cdot 0 \quad \text{at } t=0$$

$$am = C \cdot 0$$

$$A \cdot t = 0$$

$$P \cdot t = 0$$

$$ad = 0 \cdot 0 = 0$$

$$R = 0$$

$$P \cdot 0 = 0 \cdot 0 = 0$$

$$e) \frac{dy}{dx} = e^y$$

$$dy = e^y dx$$

$$\frac{dy}{1} = e^y dx$$

$$\ln(dy) = \ln(e^y dx)$$

$$\ln(dy) = y \ln e + \ln dx$$

$$\frac{dy}{dx} = e^y$$

$$\frac{dy}{dx} = e^y$$

$$d) \frac{dy}{e^y} = dx$$

$$e^{-y} dy = dx$$

$$-e^{-y} = x + c$$

$$-e^{-3} \rightarrow -e^0 = 3 + c$$

$$c = -7$$

$$-e^{-y} = -4$$

$$-y = \ln e^{-y} = 4$$

$$-y = \ln 4$$

$$y = -\ln 4$$

$$\text{defined for } x < 4$$

$$4b) \frac{dT}{dt} = k(T_e - T)$$

$$dT = dt \cdot k(T_e - T)$$

$$\frac{dT}{(T_e - T)} = k dt$$

$$\int \frac{dT}{(T_e - T)} = \int k dt$$

$$dT (T - T_e)^{-1} = -k dt$$

$$\ln |T - T_e| = -kt + c$$

$$T - T_e = e^c e^{-kt}$$

$$= A e^{-kt}$$

$$T_0 = A$$

$$T_0 = A e^{-k \cdot 0}$$

$$T_0 = A$$

$$T_0 - T_e = A$$

$$T = T_e + (T_0 - T_e) e^{-kt}$$

$$c) T = T_e + (T_0 - T_e)e^{-kt}$$

A Since $k > 0$, e^{-kt} becomes 0 as $t \rightarrow \infty$
so; $T = T_e$

$$d) T_0 = 680, T_e = 40, T(8) = 200$$

$$200 = 40 + (680 - 40)e^{-k \cdot 8}$$

$$e^{-8k} = \frac{1}{4}$$

$$-8k = -\ln 4$$

$$k = \frac{\ln 4}{8}$$

$$\therefore 50 - 40 = (680 - 40)e^{-kt}$$

$$10 = 640e^{-k}$$

$$e^{-k} = \frac{1}{64}$$

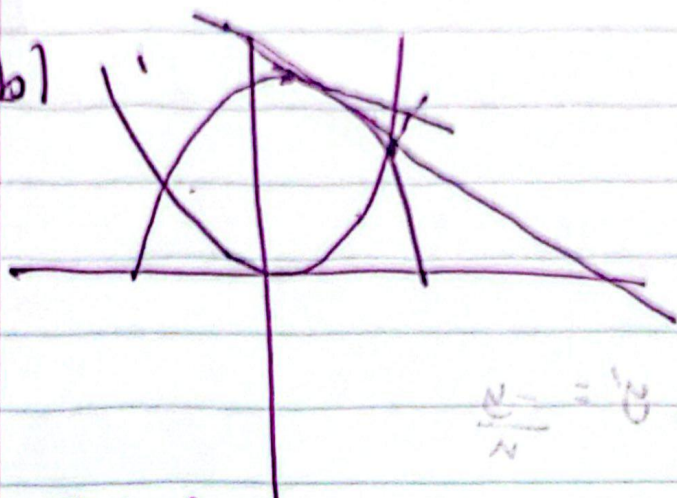
$$-k = \ln\left(\frac{1}{64}\right)$$

$$-k = -3\ln(4)$$

$$\frac{(-k)}{-8k} = \frac{-3\ln(4)}{-\ln(4)} = \frac{3T - 0T}{T - 0T}$$

$$t = 24$$

8 b)



$P = (n, y)$

$$Y - y = y' (X - n) \quad | \quad n = \text{x intercept}$$

$$-y = y' (X - n)$$

$$\frac{-y}{y'} = X - n \quad | \quad X = x - \frac{y}{y'}$$

y intercept

$$\left(n - \frac{y}{y'}, 0 \right) \quad (0, y - xy')$$

$$Y = y - xy' \quad | \quad \frac{0}{2} = 0$$

$$X = \frac{n - y}{y'}$$

$$X = \frac{1}{2}n - \frac{1}{2}\frac{y}{y'}$$

$$2n - n = -\frac{1}{2}\frac{y}{y'}$$

$$n = -\frac{1}{2}\frac{y}{y'}$$

n =

$$X = -\frac{1}{2}y^2$$

$$X = -\frac{1}{4}y^2 + c$$

$$2u = x - \frac{y}{u}$$

$$x = \frac{-y}{u}$$

$$\frac{dy}{y} = \frac{-y}{u} \quad x = \frac{-y}{u}$$

$$y' = \frac{-y}{u}$$

$$\frac{dy}{y} = -\frac{du}{u}$$

$x=$

$$[u] = 1$$

$$\ln|y| = \ln|u| + C \quad u = u - Y$$

$$y \rightarrow \ln|y| + \ln|u| = C$$

$$(u-x) \cdot u = u -$$

$$\ln|uy| = C$$

$$\frac{u-x}{u} = \frac{y}{u} = A \text{ where } A = e^C$$

$$(u-x) \cdot u = \frac{y}{u} \quad y = \frac{A}{u}$$

$$[u-x] \cdot u = y$$

$$y = \frac{e^C}{u} ; c > 0 \quad u^x - u = y$$

linear

$$\frac{u-x}{u} = x$$

$$\frac{u-x}{u} = x$$

$$\frac{u-x}{u} = x$$

$$\frac{u-x}{u} = x$$

$$\frac{u-x}{u} = x - x$$

$$\frac{u-x}{u} = x$$

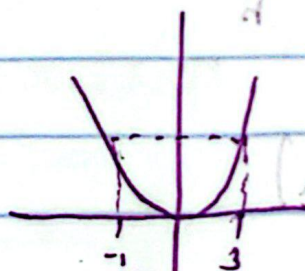
3b 2a) $3 - 5 + 7 - 9 + 11 - 13 \dots$

$$= \sum_{n=1}^{\infty} -1(2n-1) \quad \sum_{n=1}^{\infty} (-1)^{n+1}(2n+1)$$

b) $1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{n^2}$

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

3b) $\int_{-1}^3 x^2 dx$



$[-1, 0], [0, 1], [1, 2], [2, 3]$

left sum = $(-1)^2 + 1^2 + 2^2 = 6$; right sum = $0^2 + 1^2 + 2^2 + 3^2 = 14$

upper: $(-1)^2 + (1)^2 + (2)^2 + (3)^2 = 15$

lower: $(0)^2 + 0^2 + 1^2 + 2^2 = 5$

4. a) $\int_0^b x^2 dx$

$$\Delta x = \frac{b}{n}$$

$[0, \frac{b}{n}], [\frac{b}{n}, \frac{2b}{n}], [\frac{2b}{n}, \frac{3b}{n}] \dots [\frac{(n-1)b}{n}, b]$

upper - lower = $\sum [f(\text{right}) - f(\text{left})] \cdot \Delta x$

$$5. \int_0^1 e^x dx$$

$$\Delta x = \frac{1}{n}$$

$$0, \frac{1}{n}, \frac{2}{n}, \frac{3}{n}, \dots, \frac{(n-1)}{n}$$

$$\sum_{k=0}^{n-1} e^{(\frac{k}{n})} \cdot \left(\frac{1}{n}\right) = \frac{1}{n} \left[e^0 + e^{(\frac{1}{n})} + e^{(\frac{2}{n})} + \dots + e^{\frac{n-1}{n}} \right]$$

$k=1$

$$= \frac{r^n - 1}{r - 1}$$

$$= \left(\frac{e - 1}{e^{\frac{1}{n}} - 1} \right) \cdot \frac{1}{n}$$

$$= \frac{e - 1}{1}$$

$$= e - 1$$